Indexing and Query Processing (DBMS Unit 4)

An index is a data structure that speeds up retrieval of rows at the cost of extra storage and slower writes. Without an index the database performs a sequential scan, reading every row of the table.

B+ trees are the default index structure in most relational databases. All values are stored in the leaf nodes, which are linked together, so range queries such as finding students with marks between 60 and 80 are efficient.

Hash indexes support only equality lookups. They map a key through a hash function to a bucket, giving constant-time access for exact matches but no help for ranges or sorting.

A clustered index determines the physical order of rows in the table, so there can be only one per table. A non-clustered (secondary) index stores pointers to the rows and a table can have many.

GIN (generalized inverted index) is used in PostgreSQL for full-text search: it maps each word (lexeme) to the list of rows that contain it. HNSW is a graph-based approximate nearest neighbour index used by pgvector to search embeddings quickly.

The query optimizer estimates the cost of alternative plans, such as index scan versus sequential scan or nested loop join versus hash join, and picks the cheapest. EXPLAIN ANALYZE shows the chosen plan and its actual timing.